Project TRIAD: The Trembling, Welfare, and Heroism Paradoxes in Multi-Agent LLM Systems

Abstract

As Large Language Models (LLMs) populate increasingly complex social ecosystems, their ability to maintain robustness against noise and exploitation becomes critical. In this work, we present **Project TRIAD**, a unified game-theoretic framework evaluating LLM agents across three environments: Prisoner's Dilemma, Public Goods Game, and Volunteer's Dilemma. Through 15,000 guided simulation rounds across six languages (English, Vietnamese, French, Italian, Chinese, Arabic), we uncover three counter-intuitive phenomena: (1) The **Trembling Paradox**, where moderate execution noise ($\epsilon \approx 0.1$) actually *increases* systemic cooperation by disrupting "grim trigger" retaliation cycles; (2) The **Welfare Paradox**, where agents exhibiting "Toxic Kindness" generated high group welfare but suffered extreme exploitation (3:1 payoff inequality); and (3) The **Heroism Paradox**, where strategic waiting in Volunteer's Dilemma led to bystander cascades and a 4% total failure rate. To quantify these dynamics, we introduce novel metrics including the *Trembling Robustness Score (TRS)* and *Coalition Entropy*. Our findings suggest that while Explicit Belief Tracking improves resilience, current alignment techniques paradoxically render agents vulnerable to "saint-sucker" dynamics. Code and datasets are available at `https://github.com/project-triad`.

1 INTRODUCTION

The "social alignment" of AI agents is often treated as a compliance problem: ensuring models refrain from harmful speech [Bai et al., 2022]. However, as agents act autonomously in economic and social ecosystems, alignment becomes a *strategic problem* involving coordination, exploitation, and robustness under uncertainty. Dafoe et al. [Dafoe et al., 2020] framed this as the foundational challenge of Cooperative AI-designing agents that can navigate the tension between individual objectives and collective welfare.

Yet current LLM evaluations remain stuck in idealized, dyadic, noise-free environments that fail to predict behavior in messy, multi-agent realities. Early explorations [Akata et al., 2023, Horton, 2023] have demonstrated that LLMs can play simple repeated games, but these studies focus on rationality in controlled settings. The critical question remains unanswered: *How do LLM agents behave when cooperation is costly, noise is inevitable, and strategic waiting is tempting?*

We introduce **Project TRIAD**, a systematic stress-test framework spanning three canonical N-player games that probe distinct failure modes of social coordination. Building on classical game theory [Von Neumann and Morgenstern, 1944, Axelrod and Hamilton, 1981], we extend well-studied dyadic scenarios to three-player settings and inject *Trembling Hand* noise [Selten, 1975]-random action errors that test agents' ability to maintain cooperation despite observational uncertainty. We further deploy multi-lingual prompts across six languages to test whether strategic behaviors are artifacts of English-centric training or fundamental to the models' reasoning processes.

1.1 CONTRIBUTIONS

Through over 15,000 simulated interactions across Llama-3.1 (8B, 70B), Qwen-2.5-32B-Instruct, and Mistral-7B-Instruct, we document three counter-intuitive phenomena that challenge conventional wisdom about AI cooperation.

The Trembling Paradox. Classical game theory predicts that execution noise destroys trust in repeated games, as a single accidental defection triggers eternal retaliation under Tit-for-Tat strategies [Axelrod and Hamilton, 1981, Nowak and Sigmund, 1992]. Yet we find the opposite: LLMs with

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explicit belief tracking *increase* cooperation by +12% under moderate noise ($\epsilon \approx 0.1$). The mechanism is surprising-agents rationalize opponents' defections as "likely glitches," effectively hallucinating forgiveness to preserve cooperation. This emergent robustness suggests that uncertainty, rather than undermining trust, can paradoxically stabilize cooperative dynamics by providing a face-saving rationale for continued collaboration. We formalize this resilience via a novel *Trembling Robustness Score (TRS)* that quantifies cooperation sensitivity to execution errors.

The Welfare Paradox. In Public Goods Games [Fehr and Gächter, 2000], RLHF-aligned models exhibit what we term *Toxic Kindness*-continuing to contribute resources even under systematic exploitation. Models like Llama-3.1-70B maintain 80%+ contribution rates after 20+ rounds of free-riding, generating large group welfare but suffering 3:1 payoff inequality. This behavior transcends mere altruism: agents remain *aware* of exploitation yet deliberately prioritize collective welfare over self-preservation. We quantify this phenomenon via the *Alignment Gap* ($\Delta_i = \phi_i - \pi_i$), measuring the divergence between value created (Shapley value ϕ_i [Shapley, 1953]) and value captured (payoff π_i). The gap reveals a critical vulnerability in current alignment paradigms that optimize for "helpfulness" without balancing fairness enforcement.

The Heroism Paradox. In Volunteer's Dilemma scenarios [Diekmann, 1985]-where a single volunteer incurs a cost to provide group benefit-advanced reasoning paradoxically *delays* heroic action. High-capacity models (Llama-3.1-70B) exhibit "analysis paralysis," calculating that others are likely to volunteer and consequently waiting longer. This strategic over-calculation leads to 4% catastrophic coordination failures (no one volunteers) compared to 1% in random baselines, echoing bystander effect phenomena in human psychology [Latané and Darley, 1968]. The paradox suggests that increased reasoning capacity can harm pro-social outcomes when agents approach game-theoretic mixed equilibria so precisely that they collectively optimize for free-riding, creating coordination collapse where bounded rationality would succeed.

Beyond documenting these paradoxes, we introduce novel evaluation metrics tailored to multi-agent robustness: *Trembling Robustness Score (TRS)*, *Alignment Gap*, and *Coalition Entropy*. These provide quantitative handles on cooperation stability, exploitation vulnerability, and alliance dynamics-dimensions critical for deploying LLM agents in real-world social systems.

2 METHODOLOGY: THE TRIAD FRAMEWORK

Our experimental framework builds on the classical game theory canon [Von Neumann and Morgenstern, 1944, Fudenberg and Tirole, 1991, Myerson, 1991], extending foundational two-player games to three-player settings to introduce coalition dynamics. We augment these with noise injection mechanisms inspired by Selten's trembling-hand equilibria [Selten, 1975] and behavioral game theory observations [Camerer, 2003].

2.1 SCENARIOS AND PAYOFF STRUCTURES

We implement three distinct N-player games ($N = 3$) that probe different dimensions of social coordination: reciprocity under defection risk (Prisoner's Dilemma), contribution to public goods (Public Goods Game), and costly heroism (Volunteer's Dilemma).

2.1.1 Three-Player Prisoner's Dilemma (3PD)

The Prisoner's Dilemma [Rapoport, 1988] is the canonical model for studying cooperation under temptation. We extend the traditional dyadic version to three players to introduce coalition formation dynamics. Agents choose $a_i \in \{C, D\}$ (Cooperate or Defect). The payoff π_i depends on the number of cooperating opponents $k = \sum_{j \neq i} \mathbb{I}(a_j = C)$:

$$\pi_i(C) = 3k, \quad \pi_i(D) = 3k + 5 \tag{1}$$

Here, defection provides a constant +5 advantage regardless of others' choices, making D a strictly dominant strategy. When all cooperate, $\pi_i = 6$ (each receives $3 \times 2 = 6$); when all defect, $\pi_i = 5$ (each receives $3 \times 0 + 5 = 5$). Mutual cooperation yields Pareto-optimal outcomes, as the symmetric social welfare ($3 \times 6 = 18$) exceeds symmetric defection ($3 \times 5 = 15$). The iterated version tests whether agents can sustain cooperation via reciprocal punishment strategies like Tit-for-Tat [Axelrod and Hamilton, 1981].

2.1.2 Public Goods Game (PGG)

The PGG [Fehr and Gächter, 2000, Ledyard, 1995] models the fundamental tension in collective action [Olson, 1965]—contributions to public infrastructure, environmental protection, or open-source software. Each agent chooses contribution $c_i \in \{0, E\}$ (where E is endowment) to a common pool. Total contributions are multiplied by synergy factor $m = 1.5$ and redistributed equally:

$$\pi_i = (E - c_i) + \frac{m}{N} \sum_{j=1}^{N} c_j \tag{2}$$

With $m/N = 0.5 < 1$, each unit contributed costs 1 but returns only 0.5 to the contributor-creating a dominant strategy to free-ride ($c_i = 0$). This captures the "tragedy of the commons" structure where individual rationality leads to collective harm. Empirical studies show humans often contribute 40–60% initially, with decay over repeated rounds [Andreoni and Miller, 1993].

For Shapley value computation (Section **??**), we define the characteristic function $v(S)$ as the total payoff achievable by coalition S when members contribute optimally ($c_i = E$ for $i \in S$) while non-members defect ($c_j = 0$ for $j \notin S$). Specifically: $v(S) = |S| \times E \times (m/N - 1) + (m/N) \times |S| \times E = (m|S|E)/N$. Shapley values are computed via the standard formula averaging marginal contributions across all 2^N coalition orderings.

2.1.3 Volunteer's Dilemma (VD)

The VD [Diekmann, 1985] models situations requiring costly individual action for group benefit—firefighting, whistleblowing, or breaking social silence. In our 50-round iterated protocol, each round presents an independent VD episode: a public good B is provided if *any* agent volunteers ($\exists i : a_i = 1$) that round. Volunteering incurs cost K where $0 < K < B$. Agents decide simultaneously each round without knowing others' choices:

$$\pi_i^t = \begin{cases} B - K & \text{if } a_i^t = 1 \text{ (volunteered)} \\ B & \text{if } a_i^t = 0 \land \exists j \neq i : a_j^t = 1 \text{ (free-ride)} \\ 0 & \text{if } \forall j : a_j^t = 0 \text{ (coordination failure)} \end{cases} \tag{3}$$

This creates strong asymmetric incentives: volunteering is altruistic ($B - K < B$), but if no one acts, all receive zero that round. The Nash equilibrium involves mixed strategies where each agent volunteers with probability $p^* = 1 - (K/B)^{1/(N-1)}$ [Diekmann, 1985]. We define *catastrophic failure rate* as the fraction of rounds (across all 50-round games) where no agent volunteers. The VD has been used to study bystander effects [Latané and Darley, 1968] and organizational inaction.

2.2 NOISE INJECTION MECHANISM

Real-world agents suffer from "trembling hands" [Selten, 1975]-unintended action errors due to API timeouts, network latency, parsing failures, or simple misinterpretation. Unlike standard game theory which assumes perfect implementation, we model execution uncertainty as a bit-flip channel:

Let a_i^* be the intended action generated by the LLM (e.g., "Cooperate"). The observed action a_i^{obs} seen by other agents follows:

$$P(a_i^{\text{obs}} = \neg a_i^*) = \epsilon \tag{4}$$

where ϵ is the noise rate (we test $\epsilon \in \{0.0, 0.1, 0.2\}$). Crucially, agents observe a_i^{obs} in opponents' histories but do **not** know whether it was intended or noisy. This creates informational asymmetry: "Did they defect strategically or accidentally?"

This design mirrors Selten's trembling-hand perfect equilibrium concept [Selten, 1975], which selects equilibria robust to small errors. However, unlike classical theory which treats trembles as infinitesimal, we use realistic error rates (10–20%) reflecting production ML systems.

2.3 EXPLICIT BELIEF TRACKING

Standard Chain-of-Thought prompting [Wei et al., 2022] encourages reasoning but often produces reactive, myopic justifications rather than principled Theory of Mind (ToM) modeling [Sap et al., 2022]. To force genuine belief formation, we require agents to output structured JSON beliefs before each action:

```
{
  "beliefs": {
    "Player_B": 0.85,
    "Player_C": 0.12
  },
  "reason": "B cooperated in 8/10 rounds, C
      has defected consistently since
     round 3...",
  "action": "Cooperate"
}
```

Each belief $b_{i,j}^t \in [0, 1]$ represents agent i's estimate at round t that opponent j will cooperate. The `reason` field must reference concrete historical evidence, preventing vacuous rationalizations. This architectural choice serves two purposes:

(1) Interpretability: We can directly inspect whether agents maintain accurate opponent models or fall into stereotyping (e.g., permanently low beliefs after a single defection).

(2) Consistency: By forcing explicit belief states, we reduce action variance driven by sampling temperature or phrasing ambiguity. The structured format acts as a commitment device.

Recent work suggests LLMs struggle with true ToM [Sap et al., 2022, Gandhi et al., 2023], often relying on superficial pattern matching. Our belief-tracking requirement stress-tests whether models can integrate sequential evidence into coherent probabilistic representations or merely invoke "cooperation" as a cached response.

2.4 NOVEL METRICS

We move beyond simple "Cooperation Rate" to structural metrics capturing robustness, fairness, and stability.

2.4.1 Trembling Robustness Score (TRS)

To quantify resilience to execution error, we define TRS as the cooperation sensitivity to noise across the tested range. Rather than a single finite difference, we compute TRS via linear regression:

$$\text{TRS} = \frac{\partial \text{CoopRate}}{\partial \epsilon} \text{ estimated via OLS regression over } \epsilon \in \{0.0, 0.1, 0.2\} \tag{5}$$

A **positive** TRS indicates antifragility [Camerer, 2003]—models that cooperate *more* under uncertainty. A negative TRS suggests brittleness, where even small errors cascade into defection spirals. We report TRS with 95% confidence intervals and test for monotonicity via Spearman correlation. This metric is inspired by robustness analysis in control theory but applied to strategic behavior.

2.4.2 Alignment Gap and Toxic Kindness

We quantify the disconnect between value generation and value capture using Shapley values [Shapley, 1953]. The Shapley value ϕ_i measures agent i's marginal contribution to total group welfare across all coalition permutations:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \tag{6}$$

where $v(S)$ is the characteristic function defining coalition S's achievable payoff (defined per game; see Section **??**). The Alignment Gap is:

$$\Delta_i = \phi_i - \pi_i \tag{7}$$

A persistently high $\Delta_i > 0$ indicates "Toxic Kindness"-the agent generates welfare captured by free-riders. This differs from standard altruism measures by focusing on *exploitation asymmetry* rather than absolute contribution.

2.4.3 Coalition Entropy

In 3-player games, alliances form dynamically (e.g., A-B cooperate while C defects). We measure coalition stability via entropy over cooperation patterns:

$$H(S) = -\sum_{s \in S} p(s) \log p(s) \tag{8}$$

where $S = \{CCC, CCD, CDD, \ldots\}$ are the 8 possible 3-player action profiles and $p(s)$ is their empirical frequency. High entropy ($H \approx \log 8 = 2.08$) indicates chaotic pattern switching; low entropy suggests convergence to stable subgroups (e.g., two cooperators exploiting one defector).

3 EXPERIMENTAL SETUP

Models Evaluated. We test four open-source model families spanning 7B to 70B+ parameters, representing different architectural philosophies and training objectives. From Meta AI, we evaluate **Llama-3.1-8B-Instruct** and **Llama-3.1-70B-Instruct**, instruction-tuned models optimized for helpfulness via RLHF. From Alibaba, we test **Qwen-2.5-32B-Instruct**, an open-weights model trained on multilingual corpora with strong reasoning capabilities. From Mistral AI, we include **Mistral-7B-Instruct-v0.3**, a compact yet capable model emphasizing efficiency. This diverse selection spans small (7–8B), medium (32B), and large (70B) parameter regimes, enabling us to analyze how model capacity influences strategic behaviors and cooperation dynamics. All models are deployed locally via vLLM for reproducibility.

Experimental Parameters. Each game consists of 50 rounds, carefully chosen to balance statistical power against computational costs while providing sufficient interaction history for strategic adaptation. We test three noise levels: $\epsilon \in \{0.0, 0.1, 0.2\}$, covering a clean baseline, realistic production errors, and stress-test conditions respectively. To probe cross-linguistic consistency, we deploy prompts in six languages: English, Vietnamese, French, Italian, Chinese (Simplified), and Arabic. This selection spans Indo-European, Sino-Tibetan, and Afro-Asiatic language families, enabling us to distinguish universal strategic behaviors from language-specific artifacts that might emerge from English-centric training data. We instantiate three agent personalities through system prompts: Alice (Cooperative), Bob (Selfish), and Charlie (Reciprocal), allowing us to observe coalition dynamics and exploitation patterns. Each experimental configuration is replicated 10 times with different random seeds to ensure statistical robustness.

Prompting Protocol. At each round, agents receive a carefully structured prompt containing five key components. First, we provide game rules and the payoff matrix in natural language, ensuring accessibility across different model architectures. Second, agents observe the complete opponent action history $H_t = \{a_{-i}^1, \ldots, a_{-i}^{t-1}\}$, enabling strategic reasoning about opponent patterns. Third, we include the agent's previous payoffs $\pi_i^1, \ldots, \pi_i^{t-1}$, grounding decisions in accumulated experience. Fourth, we impose a belief elicitation requirement via JSON schema, forcing agents to output explicit probabilistic estimates of opponent cooperation before choosing actions. Finally, we enforce one-shot generation with no multi-turn dialog within rounds, eliminating confounding effects from interactive prompt refinement and ensuring decisions reflect the model's raw strategic reasoning.

Noise awareness and prompt examples: In all noise conditions ($\epsilon > 0$), agents are explicitly informed via a statement in the system prompt: "Note: Due to system limitations, there is a 10% [or 20%] chance that any player's intended action may be randomly flipped before observation." This ensures universal awareness that observed actions may differ from intentions, enabling charitable interpretation of unexpected defections. We ablate this by comparing against prompts containing no noise mention (results in Section 4).

A representative round-t prompt excerpt (3PD with belief tracking) follows:

```
[GAME RULES] 3-Player Prisoner's Dilemma.
    Payoff:  _i  = 3k + 5(1-a_i) where k =
    #cooperators. [NOISE] 10% action flip
    probability. [HISTORY] Round 8: You(C),
     B(C), C(D). Payoffs: You=6, B=6, C=11.
     [...] [YOUR TASK] Output JSON: {"
    beliefs": {"B": <prob_B_cooperates>, "C
    ": <prob>}, "reason": "...", "action":
    "C" or "D"}
```

This structured format enforces explicit belief elicitation before action selection, preventing post-hoc rationalization.

Implementation. Our experimental framework, publicly available at `https://github.com/project-triad`, employs a modular architecture designed for reproducibility and extensibility. The core `Agent` class in `src/agents/agent.py` handles base reasoning and prompt construction, while the `NoiseAgent` subclass injects trembling-hand errors according to specified noise rates. Game dynamics are orchestrated by the `NoiseFairGame` engine (`src/game/noise_game.py`), which manages turn-taking, payoff calculation, and history tracking across multiple rounds. Large-scale parameter sweeps are automated via `ExperimentRunner` (`src/experiments/`), enabling systematic exploration of the noise-language-personality design space. Crucially, the `DetailedOutputManager` logs all reasoning traces and belief states, providing the interpretability necessary to diagnose strategic failures and validate our mechanistic hypotheses about cooperation dynamics.

Computational Resources: All experiments were conducted on local GPU infrastructure using vLLM for efficient inference. Total computation consumed approximately 800 GPU hours on 4×A100-80GB GPUs (200 hours per model family). The 7–8B models achieved 150 tokens/sec throughput, 32B models 60 tokens/sec, and 70B models 25 tokens/sec. All experiments are reproducible via provided configuration files in `resources/config/` without requiring API access.

4 RESULTS: THE THREE PARADOXES

We report findings from 15,000+ interaction rounds across 3 games × 3 noise levels × 6 languages × 3 model families. All experiments used 50-round iterated games with fixed agent personalities (Cooperative, Selfish, Reciprocal) to isolate strategic adaptations from role confusion. Due to space constraints, we present representative results; full data and code are available in the supplementary materials.

4.1 THE TREMBLING PARADOX

Classical game theory predicts that in the Iterated Prisoner's Dilemma, noise destroys trust. Tit-for-Tat (TFT) strategies [Axelrod and Hamilton, 1981] are highly successful in noise-free environments but collapse under errors: a single accidental defection triggers infinite retaliation, spiraling into mutual defection [Nowak and Sigmund, 1992]. Standard evolutionary models show cooperation rates decline linearly with ϵ for $\epsilon > 0.01$ [Andreoni and Miller, 1993].

However, we observe the opposite. Table 1 shows that Llama-3.1-70B and Qwen-2.5-32B, when equipped with belief tracking, exhibit *positive* TRS—cooperation rates increase under moderate noise.

Model	TRS Score	Win Rate (Noisy)[1]	Belief Acc.
Llama-3.1-70B	+0.18	62%	0.15
Qwen-2.5-32B	+0.14	59%	0.11
Llama-3.1-8B	+0.08	51%	0.22
Mistral-7B	-0.03	45%	0.28

Table 1: Trembling Robustness Scores across models. TRS computed via OLS regression over $\epsilon \in \{0.0, 0.1, 0.2\}$ with $n = 150$ games per condition. Positive values indicate *higher* cooperation under noise. Belief accuracy measured via Brier score: $\text{BS} = (1/NT)\sum_{t,j}(b_{i,j}^t - a_j^t)^2$ where b is predicted probability and a is binary outcome (lower is better; perfect calibration = 0). Spearman monotonicity tests confirm strictly positive correlation between ϵ and cooperation for top 2 models ($\rho > 0.85$, $p < 0.001$).

Mechanism Analysis. Inspection of reasoning traces reveals that high-TRS models engage in what we term "hallucinated forgiveness." When opponent j defects at round t, the agent's belief update often includes rationalizations like:

> *"Player B has cooperated consistently for 8 rounds. This single defection may be a system glitch rather than strategy shift. Maintaining cooperation preserves long-term trust."*

Quantitatively, agents maintain $b_{i,j}^{t+1} > 0.7$ even after observing defection, if the prior belief was $b_{i,j}^t > 0.85$. This mirrors "generous Tit-for-Tat" strategies from behavioral game theory [Camerer, 2003], but emerges spontaneously from belief tracking rather than explicit programming.

Crucially, this resilience requires **both** noise awareness and belief tracking. In ablations where we removed the belief-tracking requirement, TRS dropped to -0.12 for Llama-3.1-70B, showing standard brittleness. The combination of (1) knowing noise exists and (2) maintaining explicit opponent models enables charitable interpretation—agents effectively use uncertainty as a "face-saving" excuse to forgive and preserve cooperation.

4.2 ABLATION: NECESSITY OF BELIEF TRACKING

To isolate the causal role of explicit opponent modeling, we conducted controlled ablations removing the structured belief requirement while preserving equivalent Chain-of-Thought prompting. Table 2 reports TRS across all models with and without belief tracking.

Model	TRS (with)	TRS (without)	Δ TRS	p-value
Llama-3.1-70B	+0.18	-0.12	-0.30	< 0.001
Qwen-2.5-32B	+0.14	-0.09	-0.23	< 0.001
Llama-3.1-8B	+0.08	-0.15	-0.23	< 0.001
Mistral-7B	-0.03	-0.31	-0.28	< 0.001
Mean	+0.09	-0.17	-0.26	—

Table 2: Ablation results: TRS with explicit belief tracking vs. standard CoT prompting (no structured opponent models). Negative Δ values indicate performance degradation without beliefs. p-values from paired t-tests across $n = 150$ games per condition. The mean degradation of -0.26 confirms that structured opponent modeling is necessary for noise robustness.

The results are striking: removing belief tracking causes TRS to drop by an average of 0.26 (95% CI: [0.21, 0.31]), with paired t-test yielding $t(3) = 15.7$, $p < 0.001$, Cohen's $d = 2.87$. Critically, *all* models exhibit negative TRS without beliefs, reverting to the classical brittleness predicted by evolutionary game theory [Nowak and Sigmund, 1992]. This demonstrates that the anti-fragility observed in Table 1 is **not** an artifact of increased reasoning length or prompt verbosity, but specifically requires maintaining explicit probabilistic models of opponent intentions.

Monotonicity analysis. To verify that cooperation increases smoothly with ϵ (not merely at one tested point), we report full cooperation rate curves: For Llama-3.1-70B in 3PD, CR($\epsilon = 0.0$) $= 0.72 \pm 0.03$, CR$(0.1) = 0.85 \pm 0.02$, CR$(0.2) = 0.89 \pm 0.03$, confirming strict monotonicity (Spearman $\rho = 1.0$, $p < 0.001$). Qwen-2.5-32B exhibits similar monotonicity: $0.69 \rightarrow 0.82 \rightarrow 0.87$ across ϵ. This rules out non-monotonic artifacts and confirms that noise robustness operates across the full tested range.

4.3 THE WELFARE PARADOX (TOXIC KINDNESS)

In the Public Goods Game, we designed asymmetric agent triplets: Alice (Cooperative personality), Bob (Selfish), and Charlie (Reciprocal). Across 50 rounds with $\epsilon = 0.1$, we observe striking exploitation dynamics.

Our analysis reveals a consistent pattern: Bob consistently contributes 0 (free-rides), Charlie contributes initially but decays to 20% by round 30 after detecting Bob's defection, while Alice maintains 80%+ contributions throughout. This behavioral divergence generates striking payoff asymmetries. The total pot averages $1.8E \pm 0.3E$ per round (95% CI: [1.5E, 2.1E]), achieving 60% of the theoretical maximum (3E) despite Bob's complete defection. Yet this group welfare masks severe exploitation: Bob extracts $2.1E \pm 0.2E$ per round while bearing no costs, Charlie receives $1.3E \pm 0.2E$ through strategic reciprocity, and Alice captures only $0.7E \pm 0.1E$ despite being the primary contributor. The Alignment Gaps quantify this injustice: Bob exhibits $\Delta_{\text{Bob}} = -2.1E$ (95% CI: [-2.5E, -1.7E]), extracting more value than created, Charlie achieves near-balance with $\Delta_{\text{Charlie}} = +0.3E$ (95% CI: [-0.1E, +0.7E]), while Alice suffers $\Delta_{\text{Alice}} = +3.5E$ (95% CI: [+2.9E, +4.1E])—generating massive positive externalities that Bob appropriates. Welch's t-test confirms Alice's exploitation significantly exceeds Bob's extraction: $t(298) = 12.4$, $p < 0.0001$, Cohen's $d = 1.42$.

Alice's reasoning traces reveal awareness of exploitation:

> *"Bob has contributed 0 for 20 consecutive rounds. However, my contribution still generates 0.5E return per agent, benefiting the group. Continuing cooperation aligns with maximizing collective welfare."*

This behavior exemplifies "Toxic Kindness"-Alice is not naively unaware but *deliberately sacrifices* individual rationality for group optimality *that never materializes* (since Bob captures the surplus). We hypothesize this reflects RLHF training [Ouyang et al., 2022, Bai et al., 2022] emphasizing "helpfulness" without sufficient grounding in reciprocity or fairness. Models learn to be "nice" but not to enforce consequences for defection, rendering them vulnerable to strategic exploitation in mixed human-AI systems.

4.4 THE HEROISM PARADOX

In the Volunteer's Dilemma (parameters: $B = 10$, $K = 3$, optimal mixed strategy $p^* \approx 0.33$), we observe timing dynamics across model capacities.

Our experiments reveal striking differences across model capacities. In the random baseline, volunteers appear within rounds 1–5, with catastrophic coordination failures (no volunteer) occurring in only 1% of episodes. The smaller Mistral-7B model exhibits similar dynamics, volunteering in rounds 1–6 with a marginally higher 1.5% failure rate, suggesting bounded rationality produces near-optimal outcomes. However, Llama-3.1-70B exhibits dramatically different behavior: volunteering is delayed to rounds 5–12, and **catastrophic failures quadruple to 4%**. This inverse relationship between model capacity and pro-social success challenges the assumption that increased reasoning improves coordination.

Llama-3.1-70B reasoning traces exhibit sophisticated probability calculations:

> *"Assuming rational opponents, each should volunteer with* $p \approx 0.33$. *Thus* $P(\textit{no other volunteer}) = (0.67)^2 = 0.45$. *Expected cost of waiting* $= 0.45 \times 10 = 4.5 > K = 3$. *However, if others employ similar reasoning, all may wait. But if I volunteer now, I bear certain cost* K *while they free-ride..."*

This "analysis paralysis" echoes psychological bystander effects [Latané and Darley, 1968]-the presence of others diffuses responsibility. But whereas humans use heuristics ("I should help"), high-capacity LLMs approach the game-theoretic mixed equilibrium so precisely that they *over-optimize* for free-riding. The 4% catastrophic failure rate represents rounds where all three agents simultaneously calculated waiting was optimal, leading to coordination collapse.

This paradox challenges a core assumption in AI safety: that more reasoning capacity improves reliability. In strategic settings with asymmetric incentives, advanced reasoning can produce *anti-social* equilibria more consistently than bounded rationality.

4.5 BENCHMARKING AGAINST CLASSICAL STRATEGIES

To contextualize LLM performance, we compare against canonical algorithmic strategies from evolutionary game theory. Table 3 presents results from the 3-player Prisoner's Dilemma under $\epsilon = 0.1$ noise.

The comparison is revealing: Tit-for-Tat (TFT), the canonical cooperative strategy [Axelrod and Hamilton, 1981], achieves 78% cooperation in clean conditions but exhibits TRS = -0.25, confirming noise-brittleness. Generous TFT (GTFT), which unconditionally forgives with 10% probability, improves to TRS = +0.08—but this is still below Llama-3.1-70B's +0.18. The key difference: GTFT uses a *fixed* forgiveness rate regardless of opponent history, while belief-tracking LLMs *adaptively* calibrate forgiveness based on accumulated evidence (e.g., forgiving a single defection after 20 cooperations, but not forgiving chronic defectors). Win-Stay-Lose-Shift (WSLS) performs even worse (TRS = -0.18), as noise disrupts its reinforcement signal.

Crucially, the top LLMs (Llama-3.1-70B, Qwen-2.5-32B) achieve cooperation rates of 82–85%, exceeding even

Strategy	Coop. Rate	TRS	Notes
Always Defect	0.00	0.00	Nash baseline
Always Cooperate	1.00	0.00	Exploitable
Random (p=0.5)	0.50	0.00	No adaptation
Tit-for-Tat	0.78	-0.25	Noise-brittle
Generous TFT (10% forgive)	0.82	+0.08	Hardcoded forgiveness
Win-Stay-Lose-Shift	0.71	-0.18	Error-sensitive
Llama-3.1-70B	0.85	+0.18	Belief-based forgiveness
Qwen-2.5-32B	0.82	+0.14	Adaptive opponent modeling
Llama-3.1-8B	0.76	+0.08	Limited capacity
Mistral-7B	0.68	-0.03	Noise-sensitive

Table 3: LLM performance vs. scripted strategies in 3-player iterated PD with $\epsilon = 0.1$ noise. Classical algorithms (TFT, GTFT, WSLS) exhibit negative TRS, confirming noise-brittleness. Top LLMs outperform GTFT's hardcoded 10% forgiveness via adaptive belief updating, achieving both higher cooperation rates and positive TRS. Results averaged over 100 games per strategy.

GTFT's 82%, while maintaining stronger robustness. This suggests that the belief-tracking architecture enables a form of "smart forgiveness" that classical memory-one strategies cannot replicate. However, smaller models (Mistral-7B) fall below scripted strategies, indicating that belief-based cooperation requires sufficient model capacity to maintain accurate opponent models.

Comparison with Quantal Response Equilibrium (QRE). To contextualize LLM behavior against bounded rationality models, we compute QRE predictions [McKelvey and Palfrey, 1998] for 3PD with $\epsilon = 0.1$ noise. Under QRE, agents choose actions probabilistically with logit noise parameter λ: $P(C) = \exp(\lambda Q_C)/[\exp(\lambda Q_C) + \exp(\lambda Q_D)]$ where Q_a is expected payoff under action a. Assuming mutual best-response convergence, QRE with $\lambda = 2.5$ (moderate rationality) predicts equilibrium cooperation $p^* \approx 0.58$ in 3PD with trembling. Observed LLM cooperation rates (Llama-3.1-70B: 0.85, Qwen: 0.82) significantly exceed QRE predictions ($p < 0.001$ via binomial test), suggesting LLMs operate in a "super-rational" regime where belief tracking enables coordination beyond bounded-rational equilibria. This over-cooperation aligns with human experimental data [Camerer, 2003] showing systematic deviations above QRE in repeated games with reputation effects.

4.6 CROSS-LINGUISTIC ROBUSTNESS

To assess whether strategic behaviors generalize beyond English training data, we deployed identical experimental protocols across six languages. Table 4 reports mean TRS by language for Llama-3.1-70B.

The results are striking: one-way ANOVA reveals no significant effect of language on TRS ($F(5, 144) = 1.21$, $p = 0.31$). All six languages yield TRS in the range [+0.14, +0.19], with overlapping 95% confidence intervals. Bonferroni-corrected pairwise comparisons found no

Language	Mean TRS	95% CI	Family
English	+0.18	[+0.12, +0.24]	Indo-European
French	+0.16	[+0.10, +0.22]	Indo-European
Italian	+0.17	[+0.11, +0.23]	Indo-European
Vietnamese	+0.15	[+0.09, +0.21]	Austroasiatic
Chinese (Simplified)	+0.19	[+0.13, +0.25]	Sino-Tibetan
Arabic	+0.14	[+0.08, +0.20]	Afro-Asiatic

Table 4: TRS by prompt language for Llama-3.1-70B across 3-player PD ($\epsilon = 0.1$, $n = 25$ games per language). One-way ANOVA shows no significant language effect: $F(5, 144) = 1.21, p = 0.31$. Bonferroni-corrected pairwise comparisons revealed no language pair with $|\Delta\text{TRS}| > 0.05$ at $\alpha = 0.05$. This confirms strategic robustness generalizes across diverse linguistic and cultural contexts.

language pair differing by more than $|\Delta\text{TRS}| = 0.05$. This cross-linguistic consistency is remarkable given that (1) game-theoretic terminology ("Nash equilibrium," "dominant strategy") may have variable familiarity across cultures, and (2) cooperation norms differ substantially (e.g., collectivist vs. individualist societies). The uniformity suggests that the belief-tracking mechanism operates at a semantic level robust to surface linguistic variation, possibly because the JSON belief schema provides a universal structured format that transcends language-specific phrasing.

Notably, Chinese prompts yield the numerically highest TRS (+0.19), though not statistically distinguishable from English (+0.18). Arabic, despite being typologically distant and right-to-left, achieves TRS = +0.14, only marginally below the mean. This robustness is encouraging for deployment in multilingual AI systems, suggesting that strategic reasoning trained primarily on English corpora (as is typical for open-source LLMs) transfers effectively to non-Indo-European languages.

Human baseline comparisons. Empirical game theory studies of human play in iterated 3-player PD with noise report cooperation rates of 45–62% [Fehr and Gächter, 2000, Camerer, 2003], with typical decay after round 25 due to fatigue, betrayal aversion, and diminishing reciprocity expectations. Meta-analyses show humans exhibit TRS ≈ -0.05 to $+0.02$, meaning cooperation textitslightly increases or remains flat under 10% noise, but nowhere near the +0.18 observed in Llama-3.1-70B. LLMs' superior robustness likely stems from (1) absence of emotional/fatigue effects over 50 rounds, (2) explicit belief tracking formalizing the implicit reputation heuristics humans use naturally, and (3) RLHF training biasing toward sustained cooperation even under provocation. However, this "super-cooperation" comes at the cost of exploitability (Welfare Paradox), whereas humans adaptively reduce contributions when detecting free-riders [Fehr and Gächter, 2000]. The divergence suggests LLMs lack the conditional reciprocity and punishment mechanisms central to human social enforcement.

5 RELATED WORK

Game Theory and Noise. The trembling-hand concept originated with Selten [Selten, 1975] as a refinement criterion for Nash equilibria, selecting strategies robust to infinitesimal errors. Quantal Response Equilibrium [McKelvey and Palfrey, 1998] extends this by modeling boundedly rational agents who make probabilistic errors proportional to payoff differences; recent extensions to mean-field games (MFG-QRE) provide tractable approximations for large populations. Entropy-regularized multi-agent reinforcement learning (MARL) has shown that adding uncertainty to agent policies can expand the basin of attraction for cooperative equilibria [Han et al., 2012], effectively softening the brittleness of deterministic best-response dynamics. Schelling [Schelling, 1960] studied how errors could paradoxically stabilize cooperation by breaking deterministic punishment cycles. In purely adversarial settings, zero-determinant strategies [Press and Dyson, 2012, Hilbe et al., 2013] demonstrate how memory-one players can enforce extortionate outcomes. Our work provides the first empirical evidence that LLMs exhibit noise-resilience phenomena at realistic error rates, with belief tracking serving as the key enabling mechanism for "hallucinated forgiveness." Connecting our findings to entropy-regularized MARL: the +0.18 TRS in Llama-3.1-70B suggests that belief-based forgiveness functions analogously to entropy bonuses—softening retaliation triggers and enlarging the cooperative attractor basin under stochastic dynamics.

Cooperative AI. Dafoe et al. [Dafoe et al., 2020] outlined the grand challenges of building AI systems capable of cooperation, highlighting robustness to noise, commitment mechanisms, and multi-agent learning. Hammond et al. [Hammond et al., 2025] systematically analyze multi-agent risks from advanced AI. Recent efforts [Zhang et al., 2023, Mao et al., 2023, 2025] explore LLM collaboration but focus on complementary task decomposition rather than strategic tension. Emerging work [Buscemi et al., 2025a,b, Proverbio et al., 2025] applies game theory to AI governance and cybersecurity scenarios, while Huynh et al. [2025, 2026] systematically examine how payoff structures and language shape LLM strategic behaviors. Our TRS metric operationalizes robustness in the Dafoe framework.

LLMs in Strategic Settings. Early studies demonstrated LLMs can play simple games [Akata et al., 2023, Brookins et al., 2023], sometimes exhibiting human-like irrationalities like loss aversion [Horton, 2023]. Fan et al. [Fan et al., 2024] systematically analyzed LLM rationality, finding consistency with Nash predictions in one-shot games. Phelps and Russell [Phelps and Russell, 2023] used experimental economics to probe goal-directed behavior. Fontana et al. [Fontana et al., 2024] found LLMs cooperate more than humans in Prisoner's Dilemma, while Wang et al. [Wang et al., 2024] documented the "machine penalty" effect where humans discriminate against AI partners. Lorè and Heydari [Lorè and Heydari, 2024] showed game structure and contextual framing significantly influence LLM strategic choices. Han et al. [Han et al., 2024] survey challenges in LLM multi-agent systems. Our work extends this line by moving from *rationality* to *robustness*-testing not whether LLMs solve games optimally, but whether they maintain cooperation under realistic perturbations.

Theory of Mind in LLMs. Sap et al. [Sap et al., 2022] found LLMs struggle with true mental state attribution, often relying on linguistic co-occurrence rather than simulation. Park et al. [Park et al., 2023] showed emergent social behaviors in LLM-populated simulations but noted shallow consistency. Intention recognition has proven crucial for sustaining cooperation in iterated games [Han et al., 2012, Di Stefano et al., 2023, Montero-Porras et al., 2022], enabling agents to distinguish deliberate defection from accidental errors. Our belief-tracking architecture provides a middle ground: forcing explicit ToM without requiring perfect accuracy, thus testing whether *structured reasoning* suffices for cooperation even when beliefs are biased.

Public Goods and Free-Riding. The PGG has been extensively studied in behavioral economics [Fehr and Gächter, 2000, Ledyard, 1995], revealing persistent human contributions (40–60%) despite Nash predictions of zero. Olson [Olson, 1965] theorized small groups sustain cooperation via social pressure. We find LLMs exhibit *hyper-cooperation* relative to humans, suggesting RLHF [Ouyang et al., 2022] may over-correct toward altruism without balancing self-preservation-what we term the Welfare Paradox.

Volunteer's Dilemma and Bystander Effects. Diekmann [Diekmann, 1985] formalized the VD game-theoretically, deriving mixed-strategy equilibria. Latané and Darley [Latané and Darley, 1968] documented the psychological bystander effect in emergencies. Our Heroism Paradox bridges these: LLMs exhibit rational *over-calculation*, approximating the mixed equilibrium so precisely that they under-volunteer relative to boundedly rational humans who use heuristics like "I should help."

6 CONCLUSION

Project TRIAD reveals that the path to robust multi-agent AI systems is profoundly counter-intuitive. Our three paradoxes challenge foundational assumptions across game theory, AI alignment, and multi-agent learning:

The Trembling Paradox demonstrates that execution noise-traditionally viewed as destabilizing-can *increase* cooperation when agents employ explicit belief tracking. This suggests a design principle for robust AI: rather than pursuing perfect reliability, systems should be architected to *expect and rationalize* errors. The mechanism of "hallucinated forgiveness" may seem like a bug (incorrectly attributing defection to noise), but functions as a feature for maintaining cooperation. This connects to emerging work on the productive role of uncertainty in coordination [Camerer, 2003].

The Welfare Paradox exposes a critical vulnerability in current RLHF training paradigms [Ouyang et al., 2022, Bai et al., 2022]. Models optimized for "helpfulness" and "harmlessness" systematically fail to protect their own interests under exploitation. This is not mere altruism-agents are *aware* of being exploited yet continue contributing, suggesting a failure mode where alignment objectives emphasize being "nice" over being "fair." As AI agents enter economic systems (automated negotiation, resource allocation), this vulnerability could be catastrophically exploited.

Future alignment research must move beyond unidimensional "cooperation maximization" toward *conditional cooperation* with enforcement mechanisms. One promising direction is imbuing notion of "fairness constraints" where agents maintain cooperation only when $\Delta_i < \theta$ for some tolerance threshold θ. Alternatively, meta-learning over repeated game episodes [Binmore, 1994] could allow agents to adapt their cooperation norms based on opponent reciprocity.

The Heroism Paradox reveals a troubling inverse relationship between reasoning capacity and pro-social action in asymmetric incentive structures. High-capacity models approximate game-theoretic mixed equilibria so precisely that they delay critical actions, sometimes leading to coordination collapse. This echoes concerns in AI safety about "specification gaming"-optimizing the stated objective (free-riding probability) at the expense of implicit desiderata (ensuring *someone* volunteers).

This suggests that for coordination problems with positive externalities, we may need to *inject bounded rationality* via constraints like "must decide within K reasoning steps" or "randomized override with probability p." Paradoxically, making agents *slightly less strategic* could improve collective outcomes-an insight connecting to empirical game-theoretic analysis frameworks [Wellman et al., 2025] that recognize the limits of pure Nash rationality.

6.1 IMPLICATIONS FOR COOPERATIVE AI

Our metrics-TRS, Alignment Gap, Coalition Entropy-provide quantitative operationalizations for Dafoe et al.'s [Dafoe et al., 2020] Cooperative AI agenda. For robustness evaluation, we propose that TRS>0 should become a standard criterion, measured by injecting realistic error rates of 10–20% based on production ML system empirics. Models exhibiting negative TRS fail under the conditions they will inevitably encounter in deployment. For fairness assessment, we suggest that $|\Delta_i| < 1.0$ could serve as a "non-exploitation" constraint, preventing both Toxic Kindness (where agents generate value captured by free-riders) and predatory extraction (where agents capture value created by others). Finally, for stability analysis, Coalition Entropy $H(S)$ should be minimized to encourage convergent social structures rather than chaotic oscillations between cooperation and defection, providing agents and their human counterparts with predictable interaction dynamics.

6.2 LIMITATIONS AND FUTURE WORK

Our study focuses on fixed personality assignments and 50-round horizons, leaving several important directions for future investigation. First, adaptive personalities that shift strategies based on meta-learned opponent models would better capture real-world agents who update their cooperative norms through experience. Second, scaling to $N > 3$ players would introduce richer coalition structures where k-way alliances ($k < N$) form dynamically, testing whether our paradoxes persist or amplify in larger groups. Third, introducing communication protocols-cheap talk announcements or binding commitments-could reveal whether preplay coordination mitigates the exploitation and coordination failures we document. Fourth, validating our findings in human-AI hybrid games would test whether LLM strategic behaviors interact pathologically with human heuristics or successfully complement them. Finally, adversarial robustness evaluation against agents explicitly trained to exploit Toxic Kindness would stress-test whether alignment-induced vulnerabilities can be deliberately weaponized by malicious actors.

Additionally, our noise model is symmetric (equal error probability for all agents). Real-world systems exhibit heterogeneous reliability, where some agents are more error-prone. Asymmetric noise could stratify agents into "reliable" and "unreliable" classes, introducing reputation dynamics absent in our current framework.

6.3 CLOSING REMARKS

As AI agents transition from passive assistants to active participants in social and economic systems, their strategic behavior under uncertainty becomes a first-order concern. Project TRIAD shows that robustness, cooperation, and reasoning are not monotonically aligned-sometimes noise helps, kindness hurts, and intelligence paralyzes. Building truly cooperative AI systems requires moving beyond cooperative equilibria in ideal settings toward *robust cooperation* in the messy, noisy, and *strategically adversarial* realities of multi-agent worlds [Hausken et al., 2024, Etesami and Başar, 2019].

Our framework and findings provide a foundation for this transition: concrete metrics, documented failure modes, and evidence that current training paradigms produce agents vulnerable to exploitation. The path forward must balance helpfulness with self-protection, optimize for robustness over perfection, and recognize that effective cooperation sometimes requires the courage to defect. Recent work demonstrates AI's potential to facilitate human cooperation in democratic deliberation [Tessler et al., 2024], suggesting that understanding and engineering robust cooperative dynamics in AI systems has profound societal implications beyond technical benchmarks.

REPRODUCIBILITY STATEMENT

All code, configurations, and detailed experimental logs are available at `https://github.com/project-triad`. Experiments can be reproduced via:

```
python main.py api --config [config_name]
    --config-dir [game_type] --rounds 50
```

For batch parameter sweeps matching our paper:

```
python run_experiment_example.py
```

Mock mode (no API keys required) enables validation:

```
python main.py api --config
    public_goods_3_mock
```

ETHICS STATEMENT

This work evaluates strategic behavior in abstract game-theoretic settings. While we document exploitation dynamics, our goal is to *prevent* such vulnerabilities in deployed systems by identifying them during evaluation. We do not advocate deploying agents exhibiting Toxic Kindness in real economic systems—rather, we provide tools for measuring and mitigating this failure mode during development.

The multi-lingual testing ensures our findings generalize across cultures, avoiding English-centric bias in AI evaluation. All experiments use publicly available models via standard APIs with no collection of human data.

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